CO-S-90C
June 22, 1970
SUPERSEDING
Int. Fed. Spec. CC-S-0090B(Navy-Ships)
October 26, 1966 and
Fed. Spec. CO-S-90A
July 11, 1957

FEDERAL SPECIFICATION

SANDERS, ELECTRIC, PORTABLE, BELT, DISK, AND OSCILLATING PAD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable electric belt sanders, disk sanders, and oscillating pad sanders.

1.1.1 Federal specification coverage. Federal specifications do not include all the types, classes, styles, and sizes of the commodity indicated by the title of this specification, but are intended to cover those most generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Sanders shall be of the following types, classes, and styles, as specified (see 6.2):

Type I - Sanders, belt.

Style 1 - Without dust collecting system.
Style 2 - With attached dust collecting system.
Style 3 - With connector for attachment to an external dust collecting system.

Type II - Sanders, disk.

Class 1 - Standard duty.
Class 2 - Heavy duty.

Type III - Sanders, oscillating pad.

Style 1 - Orbital (single grip).
Style 2 - Reciprocating (2-handles).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

FSC 5130

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For the Treatment of Communications, Electronic, and Associated Electrical Equipment).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1337 - Electric Portable Hand Tools General Suppression System Design Requirements.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids, or request for proposal shall apply:

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies shall be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

American National Standards Institute (ANSI), Inc. Standards:

B3.5 - Tolerances for Ball and Roller Bearings.
B3.6 - Boundary Dimensions for Ball and Roller Bearings.
B3.8 - Mounting Ball and Roller Bearings, Specifications For.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

The Anti-Friction Bearing Manufacturers Association (AFBMA) Standards:

Section No. 5 - Bearing Identification Code for Anti-Friction Ball and Roller Bearings.

(Application for copies should be addressed to The Anti-Friction Bearing Manufacturers Association, Inc., 60 East 42nd Street, New York, New York, 10017.)

Underwriters' Laboratories (UL), Inc. Standards For Safety:

No. 45 - Electric Tools.
No. 498 - Attachment Plugs and Receptacles.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, Long Island, N.Y. 11746; or 1655 Scott Boulevard, Santa Clara, California 95950.)

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3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of sanders which are otherwise in accordance with the requirements of this specification.

3.2 First article sample. When specified (see 6.2), the contractor shall furnish a completely fabricated portable electric sander for first article testing in accordance with section 4. The sample shall be a preproduction item or a production item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed items and shall have been made by the methods that are to be used in production of items to fulfill the terms of the contract. Approval of the sample under first article inspection shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.3 Design. The sanders shall be of the manufacturer's current design. The design shall be such as to provide convenient and safe operation. All parts of the sander shall be new, designed and constructed to the requirements specified herein. All parts subject to wear, breakage and distortion shall be easily accessible for adjustment and repair.

3.3.1 Safety devices. The sander shall be fitted with safety devices or coverings for all parts that present safety hazards. The devices shall include guards for fan belts, pulleys, chains, gears and sprockets. All guards shall be designed to provide access to the guarded parts. Safety devices, guards and ventilating openings shall be within the acceptable minimum safety requirements of UL45.

3.3.2 Lubrication. Bearings, except pre-lubricated sealed bearings, gears and moving parts shall be assembled with lubricating medium. Convenient access shall be provided to areas where the manufacturer recommends periodic lubrication or lubricant replenishment is required. External lubricant fittings shall not be provided. Lubricant shall not leak from gear casing or enter motor compartment. Parts requiring protection from dust shall be enclosed.

3.3.3 Interchangeability. All parts from the same manufacturer shall be manufactured to standards which permit interchangeability of similar parts without modifiable adjustment.

3.3.4 Material. Materials shall conform to the electrical requirements of UL45. All materials used in fabrication of sanders shall be new and of uniform quality and be produced according to standard practices of reputable manufacturers producing equipment which shall conform to this specification.

3.3.5 Conformance to standards of Underwriters' Laboratories, Inc. The contractor shall submit to the contracting agency proof that the basic sander he proposes to supply under this specification conforms to the requirements of the UL45 for Electric Tools. The label, or listing with reexamination of the UL, may be accepted as evidence that the sander conforms to this requirement. In lieu of the label, or listing with reexamination, the contractor may submit independent proof, satisfactory to the contracting agency, that sander conforms to the applicable requirements of the published standards including methods of tests of the UL noted above. Compliance with requirements of UL regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his sander. In event of difference between any of the requirements of this specification and those of UL, the requirements of this specification shall govern.

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3.4 Construction. The sander shall be complete so that when connected to the specified source of power, it can be used for its designed operation. Construction shall be free of any characteristics or defects that will prevent the sander from passing any of the examinations and tests specified in section 4.

3.4.1 Castings. All castings shall be free from defects and mismatching. In no event shall such processes as welding, peening, plugging or filling with cold solders or metallic pastes be used on castings for reclaiming any parts of the sander.

3.4.2 Surfaces. Surfaces of all castings, forgings and molded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be rounded or beveled. Surfaces shall be finished according to the manufacturer's standard practice.

3.4.3 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where those operations are included in fabrication of the original design. Those techniques shall not be employed as repair measures for defective parts.

3.4.4 Fastening devices. All screws, pins, bolts and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be permanently deformed. All threaded parts shall be in the inch system and shall conform to H28.

3.4.5 Gears and shafts. Gears shall be fabricated from suitably hardened alloy steel. The gears shall have machine cut or roll-formed teeth and shall be designed and supported to efficiently transmit imposed loads. Where worm gearing is used, the gearing may consist of a steel worm and bronze worm wheel. Shafts shall be fabricated from suitably hardened alloy steel and shall be suppported by at least two bearings to prevent deflection. Gears and shafts shall be fully enclosed.

3.4.6 Bearings. All rotating parts of the sanders shall be provided with precision quality commercially available ball or roller bearings. Armature shafts shall be mounted in ball or roller bearings. Rotating shafts other than armature shafts, shall be mounted in ball, roller or needle roller bearings. Roller or needle roller bearings which are designed primarily for radial loads shall not be used to absorb thrust. Roller or needle roller bearings may be used only when hardness of shaft supported, provision for thrust loading, sealing and shielding are operationally adequate. Steel inserts may be omitted in the mounting of bearings when sufficient contact area and a suitable fit between bearing outer diameter and housing is provided. Bearings shall be of the dustproof sealed type or shall be mounted in a dustproof housing. Ball, Roller and needle roller bearings shall conform to ANSI B3.5 for tolerance, B3.6 for dimensions, B3.8 for mounting requirements, and AFBMA Section No. 5 for identification. When a separate motor is required in the dust collecting system, (type I sanders), ball or needle roller bearings shall be furnished on the additional motor. Lubricant impregnated bronze sleeve bearings may be supplied, only on the additional motor, when specified by the procuring agency (see 6.2).

3.4.7 Housings. Housings shall be of a material of suitable characteristics to withstand stresses imposed by normal usage. Housings shall include the motor case and reduction gear, cam or belt mechanism enclosures.

3.4.8 Electrical equipment. All electrical components including motors, switches and wiring shall conform to applicable requirements of UL45 in addition to the requirements specified herein.

3.4.8.1 Motors. Motors shall be universal series wound or compensated series wound, designed to operate on either 115 volt, direct current or 115 volts (± 6 percent), single phase, 25 to 60 cycles, alternating current. Motors shall be mounted integral with the tool. The motor armature shall be balanced dynamically to insure freedom from destructive vibration.

3.4.8.2 Commutator. Sander motors may be either of "V" ring and sheet mica or molded construction. The mica shall be undercut between commutator segments. Molded commutators shall be of material that will withstand maximum rotational forces and be immune to deformation from temperatures encountered in service.

3.4.8.3 Wiring, cord, and plug cap. All wiring, flexible cord and attachment plug cap shall be in accordance with UL45, as applicable. The cord shall have an attachment plug cap and unless otherwise specified (see 6.2), shall have a minimum free length of 8 feet measured from point of entry into the sander. Unless otherwise specified (see 6.2), attachment plug cap shall conform to UL498 requirements for 2-pole, 3-wire grounding devices rated 15 amperes at 125 volts.

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3.4.8.4 Switch. Each sander shall be provided with a conveniently located switch of quick break design, (AC/DC rated). Current rating of the switch shall be not less than the rated load current of the sander. Switches for type I and II sanders shall release automatically. Switches for type II and III sanders shall have a lock for holding in "ON" position.

3.5 Electromagnetic interference suppression. When specified (see 6.2), each sander shall be electromagnetic interference suppressed in accordance with the requirements of MIL-STD-1337.

3.5.1 Additional motor. When electromagnetic interference suppression is required, it shall cover separate additional motor when furnished as part of the dust collecting system (type I sanders).

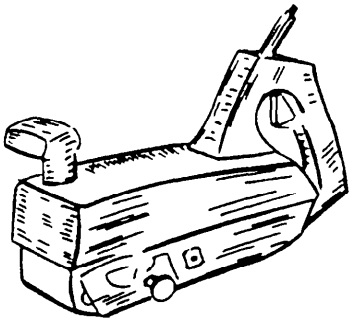
3.6 Fungus proofing. When specified (see 6.2), the electrical circuitry, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, with the following exceptions:

(a) Components or circuit elements which are inherently fungus and water resistant or which are hermetically sealed need not be treated.

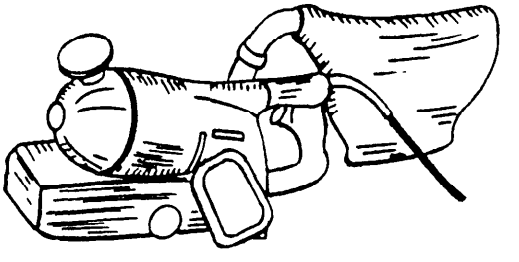
(b) Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied to give a minimum dry film thickness of 0.001 inch to component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

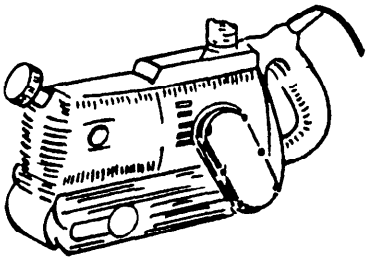
3.7 Type I, sanders, belt. Type I belt sanders shall be of three styles. Style 1 shall be without a dust collecting system, style 2 shall be equipped with an attached dust collecting system, and style 3 shall be equipped with a connector for attachment to an external dust collecting system. Type I belt sanders shall be well balanced and suitable for working next to up-rights or walls. The belt shall be easy to replace and proper belt tension shall be maintained under all working conditions. Type I belt sanders shall be similar to styles 1, 2, or 3 shown on figure 1 and shall conform to table I, as specified (see 6.2).



Style 1.



Style 2.



Style 3.

FIGURE 1. Type I, sanders, belt.

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TABLE I. Type I, sanders, belt

Item	Belt			Rated Load Current @ 115 VAC (Min.)	Width (max.)	Height without bag (max.)	Length (max.)	Net Weight	
	Width	Length (min.)	No-load speed 1/					Without dust collector	With dust collector
	Inches	Inches	Feet per minute	Amperes	Inches	Inches	Inch	Pounds	Pounds
3a	3	21	1000-1500	5-1/2	6	6.5	12	10-14	12-16
3b	3	23 3/4	1100-1700	7	7	13	22	12-16	13-18
4a	4	24	1100-1900	9	8	13	22	13-16	15-18
4b	4 1/2	26	1100-1900	10	8	13	22	20-25	22-27
4c	4	27	1100-1900	12	8	13	22	20-25	22-27

1/ The full load speed shall in no case be less than 55 percent of the rated no-load speed.

3.7.1 Components. Belt sanders shall consist essentially of housing, electric motor, switch, wiring, gears, shafts, bearings, drive mechanism, belt tracking, tensioning mechanisms, sanding shoe, striker bar, handles and standard equipment and shall conform to the requirements specified herein.

3.7.1.1 Drive mechanism. The drive mechanism shall transmit the rotary motion of the motor to the sanding belt with minimum loss of power.

3.7.1.2 Belt tracking and tensioning mechanism. Belt tracking and tensioning mechanism shall be conveniently located to insure positive alignment and proper tension of the belt and facilitate easy replacement and adjustment.

3.7.1.3 Sanding shoe. Sanding shoe shall have a resilient pad under a wearing surface of flexible sheet material.

3.7.1.4 Striker bar. Belt sanders shall be equipped with a striker bar designed to prevent belt damage to housing.

3.7.1.5 Handles. Sanders shall have a closed type handle on the rear, and a knob type handle on the front. Handles shall be conveniently located for comfortable operating control of the tool.

3.7.1.6 Standard equipment. Belt sanders shall be complete with abrasive belts and extra parts as normally furnished commercially.

3.7.1.7 Dust collecting system (styles 2 and 3). Style 2 sanders shall be equipped with a dust-collecting system that consists of a fan, a vacuum intake duct, and a bag or container into which the dust shall be deposited. The dust bag or container shall be so arranged that the accumulated dust may be conveniently removed. At the option of the contractor, the dust collector may have a separate motor which shall be operated by the same switch as the sander. Style 3 sanders shall be provided with a means of connecting the sander to an external dust-collecting system.

3.8 Type II, sanders, disk. Type II disk sanders shall be similar to figure 2 and shall conform to table II for the class and size specified (see 6.2). Class 1 sanders shall be standard duty. Class 2 sanders shall be suitable for heavy duty continuous sanding operations.

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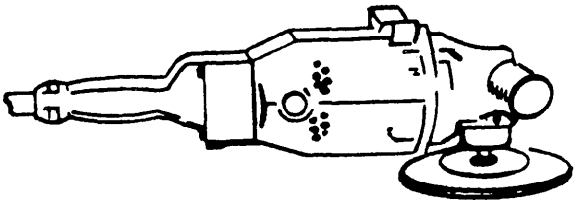


FIGURE 2. Type II, sander, disk.

TABLE II. Type II, sanders, disk

Class	Nominal Size	Disk size (dia.)	Rated HP 1/ (min.)	No-load speed 2/ (rpm)	Length without pad (max.)	Net Weight without pad or cable (max.)	Spindle thread dim. RH
	Inches	Inches			Inches	Pounds	
1	7	7	1/2	3500-5500	19	18	5/8-11
2a	7	7	3/4	3800-6000	21	20	5/8-11
2b	9	9-1/8	3/4	3800-5000	23	20	5/8-11

1/ Horsepower shall be taken at disk end of spindle, with speed not less than the full-load minimum r.p.m.

2/ The full-load speed shall in no case be less than 55 percent of the rated no-load speed.

3.3.1 Components. Disk sanders shall consist essentially of housings, electric motor, switch, wiring, gears, spindle shaft, bearings, handles and standard equipment and shall conform to the requirements specified herein.

3.3.1.1 Spindle. The spindle shall be suitably hardened alloy or forged steel and shall be supported by at least two bearings. Outer spindle bearings shall be ball or roller or a combination ball and roller designed to withstand the radial thrust loads imposed. Means shall be provided for locking or holding spindle to facilitate changing abrasive disks and backing pads. Spindle thread size shall conform to table II.

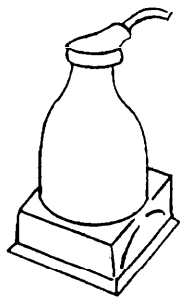
3.3.1.2 Handles. Each disk sander shall have an end handle and a reversible side handle designed for mounting either right or left of spindle. The longitudinal axis of handles shall be located to facilitate ease of handling and operation of the sander under service conditions.

3.3.1.3 Backing pad. Disk sanders shall be equipped with a semi-flexible pad for backing the abrasive disk. The pad shall be of a material that will withstand the maximum centrifugal force developed and will not deteriorate from heat created by sanding operations.

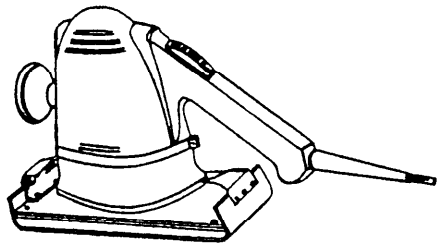
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3.8.1.4 Tool rest. Disk sanders shall be equipped with a tool rest of suitable resilient material fastened to the housing opposite the sanding disk to enable resting the tool without injury to the sander spindle and disk assembly. When specified (see 6.2), class 2 disk sanders shall have provision for suspension eyebolt mounting.

3.9 Type III, sanders, oscillating pad. Type II oscillating pad sanders shall be of two styles. Style 1 (single grip) shall have an orbital motion with not less than 4,000 cycles at no-load. Style 2 (2-handles) shall have a reciprocating motion with not less than 3,000 cycles at no-load. They shall conform to table III for size and shall be similar to figure 3 for the style,as specified (see 6.2).



Style 1.



Style 2.

FIGURE 3. Type III, sanders, oscillating pad.

TABLE III. Type III, sanders, oscillating pad

Size	Weight less cable (max.)	Test Weight	Abrasive area (min.)	Amplitude of oscillations		Work factor (min.)
				Minimum	Maximum	
	Pounds	Pounds	Square inches	Inch	Inch	
1	6	10	18	0.078	0.375	225
2	7-1/2	10	25	0.125	0.500	300
3	8-1/2	10	40	0.125	0.500	600

3.9.1 Components. Oscillating sanders shall consist essentially of housings, electric motor, switch, wiring, gears, shafts and bearings, motion transfer mechanism, backing pad, abrasive holding device, handles and standard equipment and shall conform to the requirements specified herein.

3.9.1.1 Motion transfer mechanism. When specified (see 6.2), style 2 sanders shall have a motion transfer mechanism to transfer motor rotation to orbital motion of the sanding pad. Moving elements of the mechanism shall be mounted in ball bearings designed to withstand the imposed radial thrust loads. Motive power shall be transmitted to the pad by belt, gear train, cam, direct drive or other means.

3.9.1.2 Abrasive holding device. Each sander shall be furnished with a permanently attached abrasive holding device designed to quickly remove and securely replace abrasive paper.

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3.9.1.3 Handles. Sanders designed for two-hand operation shall have a straight or pistol grip handle and an auxiliary knob handle designed for a comfortable grip and ease of operation. Sanders designed for one-hand operation shall have a functional single grip.

3.9.1.4 Standard equipment. Sanders shall be equipped with a backing pad of felt, rubber, or other durable resilient material attached to a metal plate.

3.10 Identification plate. A suitable identification plate, permanently and legibly marked to include the following information shall be securely attached to the sander:

(a) Manufacturer's name.
(b) Manufacturer's model or serial number.
(c) Voltage.
(d) Rated-load, amperes.
(e) No-load surface feet per minute for type I sanders, no-load revolutions per minute for type II sanders.
(f) Frequency orbits per minute for type III sanders.

3.11 Handbook of instruction. The manufacturer shall furnish one copy of his regular instruction book and parts list with each sander. The book shall include all illustrations, instructions and necessary data for safe operation, maintenance, adjustment and repair. The parts list shall include the manufacturer's part number and ordering instructions.

3.12 Standard equipment. In addition to standard equipment specified herein, each sander shall be furnished with such maintenance tools, lubricant, abrasive paper, and extra parts as normally furnished commercially. Additional accessory equipment shall be, as specified (see 6.2).

3.13 Workmanship. Workmanship of the sander and accessories shall be of the quality prevailing among manufacturers normally producing sanders of the types specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except, as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the sanders shall be classified as follows:

(a) First article inspection.
(b) Quality conformance inspection.

4.3 First article inspection. First article inspection shall be applied to the sample submitted in accordance with 3.2. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5.1 and all tests in 4.6. Failure of the sample to pass the examination or one or more first article tests shall be cause for rejection.

4.4 Sampling for quality conformance inspection.

4.4.1 Lot. All sanders of the same type, class and style presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.4.2 Sample. A random sample shall be selected from each lot in conformance with inspection level II of MIL-STD-105.

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4.5 Quality conformance inspection. Quality conformance inspection shall be performed on each sample sander selected in accordance with 4.4.2. Quality conformance inspection shall consist of the following:

(a) Examination (4.5.1).
(b) Tests (4.6.1 through 4.6.3).
(c) Inspection of preservation, packaging, packing, and marking for shipment and storage (4.7).

4.5.1 Examination. Each sander selected in accordance with 4.4.2 shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with table IV. The acceptable quality level (AQL) shall be as specified in table IV. Nonconforming sanders shall not be offered for delivery. If the nonconforming sanders in any one sample exceed the acceptance number for that sample, the lot represented shall be rejected.

TABLE IV. Classification of defects

Categories	Defects
Critical:	AQL 0.10 PERCENT DEFECTIVE
001	Not safe from electric shock or fire hazard.
002	Moving parts (hazardous to operator) improperly guarded.
Major:	AQL 2.5 PERCENT DEFECTIVE
101	Sander not of type, class or style specified, as applicable.
102	Component parts missing, damaged, defective or not as specified.
103	Parts not interchangeable.
Minor:	AQL 4 PERCENT DEFECTIVE
201	Dimensions and weight not as specified.
202	Identification, nameplate illegible or not permanent.
203	Improper workmanship.

4.6 Tests.

4.6.1 Electromagnetic interference test certification. When electromagnetic interference suppression is required, the manufacturer shall provide certification that the suppression network in the sanders offered is identical to that of the sanders previously tested and approved.

4.6.2 Motor tests. Each sample sander selected in accordance with 4.3.2 shall be tested to determine compliance with UL45, as modified by MIL-STD-1337, for industrially rated motors, for:

(a) Continuity of grounding connections.
(b) Dielectric strength.
(c) Current input (no-load).

4.6.3 No-load tests. Each sample sander selected shall be operated thirty minutes at no-load. During the test, the sander shall be checked for switch operation and loss of lubricant.

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4.6.4 Rated-load test. Each sample sander submitted for preproduction testing shall comply with the no-load requirements of 4.6.3. After the no-load tests, the sander shall be operated at rated-load. When operating temperatures are constant (thermal equilibrium at rated-load current, frequency and voltage) the sanders shall be operated at rated-load for one hour. Sanders shall be tested for:

(a) Temperature rise.
(b) Current Input.
(c) Dielectric strength and insulation resistance, in accordance with UL45 as modified by MIL-STD-1337.

4.6.4.1 Test examination. After tests in 4.6.4 have been completed, operational components (including motor bearings and gears) shall be examined for wear, insecure fastening, and evidence of leakage. Abrasive belt tracking and tension, drive belts or chain tension, if applicable shall be checked for adequacy. Evidence of off balance or expansion of backing pad on disk sanders shall be ascertained, if any.

4.6.5 Work factor test. Type III sanders shall show a work value in conformance with table III when calculated as follows:

4.6.5.1 Work factor. The work factor is a measure of performance and is calculated by the following formula:

$$\frac{a\ d\ n\ w}{1{,}728} = \text{work factor}$$

Where:

a = abrasive contact area in square inches.
d = distance, inches, of travel for one complete cycle of oscillation. For reciprocating motion sanders, the distance will be the amplitude times 2. For orbital motion sanders, the distance will be the perimeter of the orbit.

n = number of oscillations per minute, using the test weight as specified in table III, and using the following abrasive against a steel surface: No. 60 grit aluminum oxide, closed coating, medium weight cloth backing.

w = test weight

Example: Assume a 7 pound reciprocating electric sander will be tested. The 3 pounds shall be added to bring the test weight to 10 pounds. Then further assume that by measurement it is found that the contact area is 23 square inches, the oscillation under load is found to be 1,800 cycles per minute and amplitude is 5/8 inch.

Then a = 23, d = 1.25, n = 1,800, and w = 10
and work factor $= \frac{23 \times 1.25 \times 1800 \times 10}{1{,}728} = 300$

4.6.5.2 For definitions of abrasive contact area, oscillations, amplitude and test weight (see 6.3).

4.7 Inspection of preservation, packaging, and marking for shipment and storage. Preservation, packaging, packing, and marking for shipment and storage shall be inspected to determine conformance with the documents referenced in section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40 for Sanders, Portable, Electric.

6. NOTES

6.1 Intended use. Type I, styles 1, 2 and 3 sanders are intended for fast, heavy sanding and straight-line finishing of wood and other materials. Type II sanders are intended for smoothing surfaces, buffing and polishing of metal and other materials. Type II, class 1 sanders shall be suitable for medium duty and intermittent heavy duty sanding operations. Type II, class 2 sanders are intended for heavy duty continuous production sanding operations. Type III sanders, style 1, orbital and style 2, reciprocating motion, are intended for finish sanding of wood and other materials.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style and size (see 1.2, 3.7, 3.8, and 3.9).
(c) First article sample, when required (see 3.2).
(d) Lubricant impregnated bronze sleeve bearing, when required (see 3.4.5).
(e) Cord length or attachment plug if different (see 3.4.8.3).
(f) Electromagnetic interference suppression, if required (see 3.5).
(g) Moisture-and-fungus-treatment, if required (see 3.6).
(h) Eyebolt mounting, type II, class 2 sanders, when required (see 3.8.1.4).
(i) When motion transfer mechanism is required (see 3.9.1.1).
(j) Accessory equipment required (see 3.12).
(k) First article tests, if different (see 4.3).

6.3 Definitions. For the purpose of this specification, the following definitions apply:

(a) Abrasive contact area. The area of actual contact between the abrasive and the work when the machine is at rest and with the test weight applied.

(b) Oscillations. The number of complete cycles per minute when the machine is operating with the test weight applied. In the case of orbital motion sanders, a cycle is the describing of one complete orbit. In the case of reciprocating sanders, a cycle is one complete back and forth movement.

(c) Amplitude. The distance of travel in one direction for reciprocating motion sanders; and the maximum diameter of the orbit for orbital motion sanders.

(d) Test weight. The test weight shall consist of the actual weight of any particular sander less cable, plus an extra weight added to the top or handles of the sander to bring the total up to the test weight indicated in table III. For example, a 7 pound sander would have 3 pounds added in order to build up to the test weight.

6.4 First article.

6.4.1 Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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MILITARY INTEREST:

Custodians:

Army - GL
Navy - SH
Air Force - 84

Review activities:

Army - EL, GL
Navy - SH
Air Force - 84
DSAH - GS

User activities:

Army - ME
Navy - MC, YD

Preparing activity:

Navy - SH

CIVIL AGENCY INTEREST:

GSA-FSS
VA-DMS

Project No. (5130-0054)

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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